Blake Grudzien

San Francisco, CA — 707-331-9957 — blakegrudzien@gmail.com — blakegrudzien.com — LinkedIn — Github

Experience

Level Home May 2025 – September 2025
Software Engineer Intern | *Case study* *Redwood City, CA*

- Built the Go service behind a rebuilt device telemetry pipeline on AWS (Lambda, SQS, S3, Athena), cutting data loss from over 30% to under 1% across millions of smart lock telemetry events, and presented the architecture to 40+ people across the company
- Replaced silent deletion of unparseable files with a dead-letter queue, so the firmware team could ship a parser fix and clear the entire backlog with one redrive instead of a cross-team coordination cycle
- Chose SQS over EventBridge after finding fan-out irrelevant for a single-consumer path, and kept the native DLQ over a custom one, trading per-error retry control for zero custom code and one-command reprocessing
- Migrated telemetry storage from row-based JSON to Parquet via Apache Arrow so Athena scans only queried columns, choosing Arrow over DuckDB, whose CGo requirement would have forced changes across the service
- Built the service's test suite and CI pipeline, covering the full path from a raw telemetry file to queryable Parquet output
- Prototyped a Slack bot alerting on every parse failure, then narrowed it to a first-failure alert and a daily dead-letter count after the firmware team said knowing files failed was enough

Brooksee June 2024 – October 2024
Software Engineer Intern | *Case study* *Salt Lake City, UT (Hybrid)*

- Built a Garmin Connect IQ data field (Monkey C) and Kotlin Android app to stream live pace, distance, placement, and gun-time gaps to runners' watches for races of 5,000–10,000 participants, tested against a real-time race simulation
- Computed the gap to the runner ahead, which the data never stored, by comparing timestamped crossings at each timing point, and delivered it through a Firestore listener on Android bridged to the watch over Bluetooth
- Derived each runner's elapsed time from the race's absolute start timestamp rather than a local timer, so runners crossing the line at irregular intervals after the gun would still see correct official time

Education

Emory University Atlanta, GA
B.S. Computer Science, Minor in Mathematics | *GPA: 3.65/4.00* *May 2026*

Projects

Chess Scholar (Live | Github | Case study) | ***Python, Claude API, PostgreSQL/pgvector*** July 2026 – Present

- Built a hybrid retrieval system over 127,435 games and 10.2M moves (1610–2023), routing questions across deterministic SQL, pgvector search over 468,862 embedded annotation chunks, and pooled Stockfish subprocesses through Claude's native tool-calling rather than LangChain or a hand-rolled router
- Required engine grounding before any model claim about move quality, constraining the LLM to attribute answers to retrieved stats, prose, or evaluations rather than its own judgment
- Trained a gradient-boosted classifier (0.887 cross-validated ROC-AUC) to filter scraped study material before surfacing it, and covered the system with 400 tests in CI on Python 3.11 and 3.12, run against a real Postgres/pgvector instance
- Wrapped three paid APIs and PostgreSQL in explicit error handling, with an 8 req/min session cap to stop runaway usage

Bird Strike Analysis (Live | Github) | ***SvelteKit, D3.js, TypeScript*** Feb 2026 – May 2026

- Built an interactive data-driven article analyzing 300k+ FAA bird strike records, featuring animated geographic maps, linked charts, and a species heatmap, deployed via GitHub Actions

CHOA Chatbot | ***Python, React, MongoDB*** Jan 2024 – May 2024

- Built an AI-powered chatbot for Children's Healthcare of Atlanta in a 7-person team, training resident doctors on pediatric airway management protocols for respiratory emergencies

Technical Skills

Languages: Python, Go, SQL, TypeScript, JavaScript, Kotlin, Java, R
Frameworks & Libraries: React, SvelteKit, Streamlit, Scikit-Learn, pandas, NumPy, pytest, D3.js, Playwright
Cloud & Data: AWS (Lambda, SQS, S3, Athena, Kinesis), Parquet, PostgreSQL/pgvector, Firestore, Cloudflare Workers
Tools & Other: Git, GitHub Actions, Docker, Linux, REST APIs, RAG, vector embeddings, LLM tool-calling

Leadership

Vice President Oct 2023 – Sep 2024
Alpha Tau Omega Fraternity, Emory University (100+ members) *Atlanta, GA*

- Led chapter to 3rd highest GPA of 136 national ATO chapters; headed the disciplinary process, mediating conflicts

Additional

Chess: 2209 chess.com elo (99.9 percentile) | **Languages**: Intermediate Spanish | **Interests**: Tennis, Rock Climbing, Soccer